Relay Control Board V3

Design Development and Functional Testing

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Design target: 250 V AC, 5 A load, with a 3–5 V digital control input

Abstract

This report documents the design development of the Relay Control Board V3, a printed circuit board (PCB) that allows a low-voltage digital signal (3–5 V) to switch an AC load through an electromechanical relay. The project began as a protoboard circuit for controlling a water-heating element, with the intended application of an ESP32 providing wireless ON/OFF control and a PID controller holding the water at 75 °C. Across three board revisions, the design was changed to correct narrow copper traces and an excessive ground plane, to move from FR-2 to FR-4 substrate, to increase the physical separation between the AC and low-voltage sections, and to add a 5 A fuse, with a 3 mm trace width in the AC section and a 1.8 mm clearance between high-voltage conductors. Version 3 was manufactured and tested with a connected load, and the board is reported to be working. The test record is, however, incomplete: voltage, current, load type, duration and fuse, trace-width and clearance data have not been documented. The report therefore presents Version 3 as a functioning prototype whose 250 V AC, 5 A rating still requires documented verification, and it lists the evidence needed to close that gap.

Keywords: relay control board; PCB design; optocoupler isolation; mains switching; FR-4; fuse protection; ESP32; PID temperature control

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1. Project Overview

The Relay Control Board V3 is a printed circuit board designed to allow a low-voltage digital signal to control an AC load through a relay. The project originally began as a protoboard-based circuit intended to control a water-heating element.

The planned application involved an ESP32 providing wireless ON/OFF control and a PID controller maintaining the water temperature at a target of 75 °C.

The design evolved through three versions. The third revision addresses PCB trace-width limitations, ground-plane layout, separation between the AC and low-voltage sections, and circuit protection.

This report covers the relay board itself: its requirements, architecture, revision history and the state of its testing. The ESP32 temperature-control application is treated as a separate system (Section 6.3).

2. Design Requirements

The board was designed around the specifications in Table 1.

Table 1. Design specifications of the Relay Control Board V3.

Parameter	Design specification
AC voltage	250 V AC
Load current	Up to 5 A
Digital input	3–5 V
Power supply input	12 V, 1 A
Switching element	Electromechanical relay
PCB substrate	FR-4
Protection	5 A fuse
Trace width (AC section)	3 mm
Clearance between high-voltage conductors	1.8 mm

Note on interpretation. These values describe the design specifications. They should not be interpreted as proof that every operating condition has been independently validated.

3. System Architecture

3.1 Functional blocks

The circuit includes a digital input, an optocoupler, a transistor relay driver, a relay, and AC input/output terminals.

The optocoupler is intended to provide electrical isolation between the digital control signal and the relay-driving circuitry. The transistor drives the relay coil, while the relay contacts provide normally open (NO) and normally closed (NC) switching options.

The board includes two principal functional areas: the low-voltage control circuitry and the AC switching section. Their physical arrangement was revised in Version 3 to increase separation between these areas.

3.2 Circuit description

Figure 1 shows the circuit schematic and Table 2 lists its components. As drawn, the digital signal drives the input LED of the PC817 optocoupler (U1) through R1. The optocoupler output is pulled up to +12 V by R2 and is connected through R3 to the base of the PNP transistor Q1, which supplies the relay coil (K2) from the +12 V rail. D1 is connected across the coil, and a separate LED (D2) with its series resistor R4 is connected between +12 V and ground. The relay contacts and the 5 A fuse (F1) connect to the AC_IN1, NC1 and NO1 terminals.

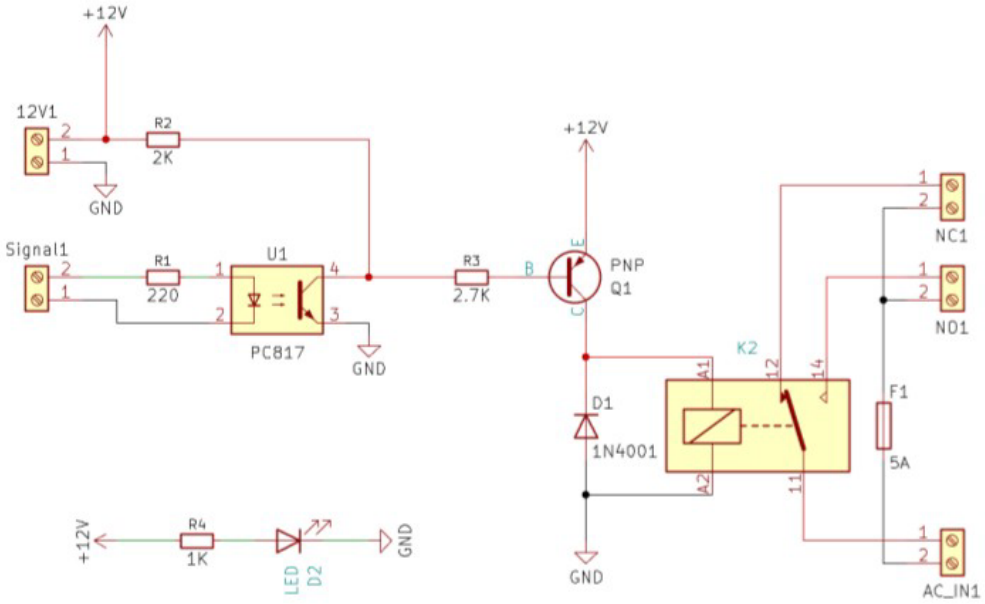


Figure 1. Circuit schematic of the Relay Control Board V3.

Table 2. Components identified in the schematic (Figure 1).

Ref.	Component / value	Role in the circuit (as drawn)
U1	PC817 optocoupler	Separates the control signal from the relay-driving circuitry
R1	220 Ω	Series resistor at the optocoupler input (Signal1)
R2	2 kΩ	Connects the optocoupler output node to +12 V
R3	2.7 kΩ	Connects the optocoupler output node to the base of Q1
Q1	PNP transistor	Relay coil driver, supplied from +12 V
D1	1N4001 diode	Connected across the relay coil
K2	Relay, 12 V DC coil	Switches the AC load; provides NO and NC contacts
F1	5 A fuse	Circuit protection in the AC section
D2, R4	LED, 1 kΩ	LED branch between +12 V and ground
12V1	Terminal block	12 V supply input
Signal1	Terminal block	3–5 V digital control input
AC_IN1, NC1, NO1	Terminal blocks	AC input, normally closed output, normally open output

4. Design Development

4.1 Previous revisions

During development, the earlier board versions presented two PCB layout problems:

- Copper traces were too narrow for the intended design requirements.
- The ground-plane layout covered an excessive portion of the board.

These issues motivated a revision of the PCB layout and routing.

4.2 Version 3 improvements

PCB substrate. The board was manufactured using FR-4 instead of the FR-2 material used previously.

Ground-plane layout. The ground-plane arrangement was modified to address the layout problems identified in the earlier version. This change describes a PCB layout modification; the exact electrical connectivity of the grounds must be established from the schematic and layout.

AC and low-voltage separation. The AC section was separated from the low-voltage circuitry to provide greater physical clearance between the sections.

Trace width. A trace width of 3 mm was used for the high-current AC-section traces, addressing the narrow-trace problem of the earlier versions (Figure 2a). The copper thickness and the current-carrying calculation supporting this width have not yet been documented.

High-voltage clearance. A clearance of 1.8 mm was specified between high-voltage conductors. This is a design value; the clearances of the manufactured board have not been measured, and

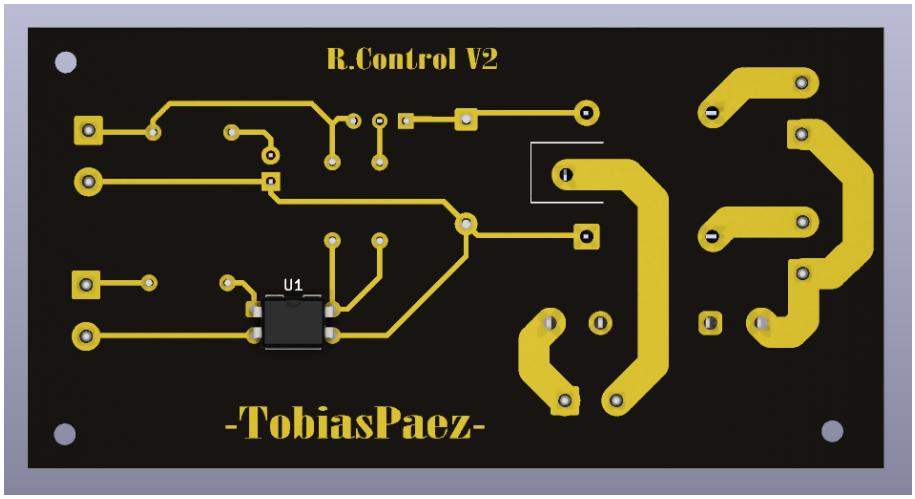
their suitability for 250 V AC remains to be verified.

Fuse protection. A 5 A fuse was added to the revised design. Its interrupting capacity, voltage rating, time-current characteristics, and suitability for the circuit require verification against the selected component's specifications and the intended application.

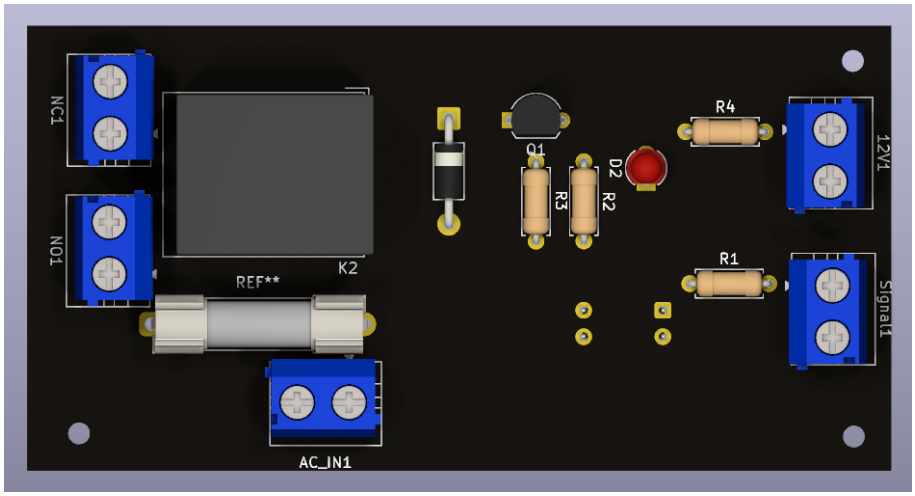
Table 3. Summary of problems identified and Version 3 responses.

Problem or requirement	Version 3 response	Verification status
Traces too narrow for the design requirements	PCB layout and routing revised	3 mm trace width specified; calculations and copper specification not yet documented
Ground plane covered too much of the board	Ground-plane arrangement modified	Ground connectivity to be established from schematic and layout
Need for AC / low-voltage separation	AC section physically separated from low-voltage circuitry	1.8 mm clearance specified; measured clearances and insulation requirements not yet documented
Circuit protection	5 A fuse added	Ratings and suitability to be verified
Substrate	FR-4 used instead of FR-2	Board manufactured and tested with a load

Figure 2 shows the PCB layout as rendered from the design files, with the copper layer in (a) and the component side in (b). Figure 3 shows the manufactured FR-4 board, in which the relay, fuse holder, indicator LED and screw terminals are visible.



(a)



(b)

Figure 2. PCB layout: (a) copper layer; (b) component side.

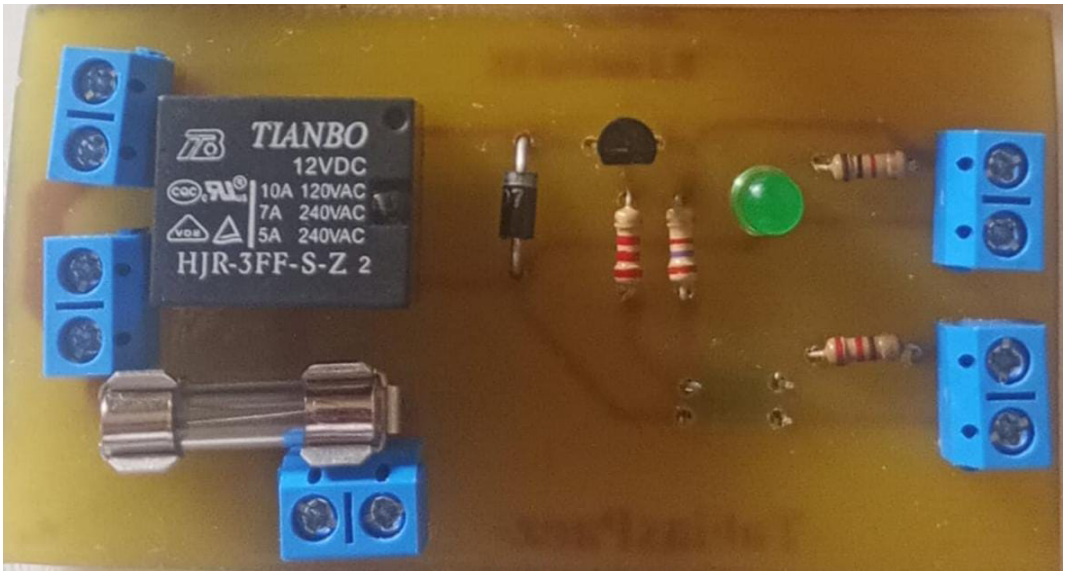


Figure 3. Manufactured FR-4 board.

5. Functional Testing

Version 3 was physically manufactured and tested with a connected load. The board is reported to be working.

However, detailed test observations, measurements, and operating conditions have not been recorded in the available documentation.

The current evidence therefore supports reporting that a load-connected functional test was performed. It does not establish the exact voltage and current reached during testing or demonstrate continuous operation at the full design target of 250 V AC and 5 A.

Table 4. Evidence status of the functional test.

Aspect	Status
Version 3 manufactured on FR-4	Established (Figure 3)
Tested with a connected load; board reported working	Reported
Test supply voltage and load current	Not recorded
Load type and duration of operation	Not recorded
Relay switching behavior and abnormalities	Not recorded
Continuous operation at 250 V AC and 5 A	Not demonstrated

The following verification items remain to be documented:

- Test supply voltage and load current.
- Load type and duration of operation.
- Relay switching behavior and any observed abnormalities.
- Fuse specifications and protection suitability.
- PCB trace-width calculations and copper specifications (design width: 3 mm).
- Measured clearances (design value: 1.8 mm) and applicable insulation requirements.
- Verification of the intended isolation between the control and power circuits.

Passing PCB design-rule checks, if performed, would not by itself establish mains-voltage safety.

> **Safety notice.** The board is designed to switch mains-level voltage. Until the verification items above are documented, it should be handled and used only with appropriate precautions for mains voltage.

6. Limitations and Future Improvements

6.1 Documentation and verification

The next development step is to document the electrical characteristics of the selected components and the calculations supporting the intended operating conditions.

Further verification should establish whether the PCB routing, fuse selection, physical clearances, and isolation arrangements are appropriate for the intended voltage and current. In particular, the 3 mm trace width should be supported by a current calculation that states the copper thickness and permitted temperature rise, and the 1.8 mm clearance should be checked against the applicable insulation standard, including the creepage distance along the board surface as well as the air clearance.

6.2 Additional points noted during this revision of the report

- **Revision labelling.** The silkscreen text in the layout render (Figure 2a) reads “R.Control V2”, while this report describes Version 3. The label and the revision number should be reconciled. The fuse footprint in Figure 2b also shows the placeholder reference “REF**” instead of F1.
- **Conductor assignment.** In the schematic the relay contacts are placed in one AC conductor and the fuse in the other. The line and neutral connections should be marked on the board and confirmed, so that the intended protection and switching arrangement is clear to the user.

6.3 ESP32 temperature-control application

The original ESP32-based temperature-control application should also be evaluated separately from the relay board itself. Its PID performance and temperature stability have not been established by the information currently available.

7. Conclusion

The Relay Control Board V3 is the third iteration of a relay-control PCB developed to address problems identified during earlier design stages.

The revised board was manufactured using FR-4 and incorporates a modified ground-plane layout, greater physical separation between the AC and low-voltage sections, a 3 mm trace width in the AC section, a 1.8 mm high-voltage clearance, and a 5 A fuse. Functional testing with a connected load has been reported.

Although the board is reported to be working, additional documented measurements and electrical safety verification are required to substantiate its intended 250 V AC and 5 A operating conditions.

The project demonstrates an iterative hardware-development process in which problems identified in earlier PCB versions informed subsequent design changes.